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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING /
SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2022-23

Programme: B.Tech / MBA Tech (All Stream) / B.Tech Integrated (Civil)
Year: II/III/IV Semester:III/V/VII

Comp Sem3

Subject: Management Accounting for Engineers

Date: 25 November 2022

Marks: 100
Time: 10.00 am - 1.00 pm
Durations: 3 (Hrs)
No. of Pages: 05

Final Examination (2022-23)/ Re-Examination (2021-22)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) **In all ___5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

<table>
<tr><td>Q.1</td><td></td><td>Answer briefly and show detailed calculations.</td><td>20</td></tr>
<tr><td>CO- 1; SO- ; BL- Remember</td><td>a</td><td>What are the essentials of a Good Costing System?</td><td>5</td></tr>
<tr><td>CO-4 ; SO- ; BL- Evaluate</td><td>b</td><td>Radhika Pvt. Ltd. Produces a product "R". The unit selling price and variable cost per unit is Rs.1000 and Rs.750 respectively. Fixed cost is Rs.25,50,000 p.a.
a. Calculate Break-even point of the company.
b. Also, determine the expected sales if desired profit is Rs.7,50,000.</td><td>5</td></tr>
<tr><td>CO- 4 ; SO- ; BL- Evaluate</td><td>c</td><td>From the following information compute, Material Cost Variances, Material Price Variance and Material Usage Variance.
<table>
<tr><th>Material</th><th colspan="3">Standard cost for 1 unit</th><th colspan="3">Actual cost for 10 units</th></tr>
<tr><th></th><th>Quantity</th><th>Rate</th><th>Amount</th><th>Quantity</th><th>Rate</th><th>Amount</th></tr>
<tr><td>X</td><td>60</td><td>15</td><td>900</td><td>640</td><td>17.5</td><td>11200</td></tr>
<tr><td>Y</td><td>80</td><td>20</td><td>1600</td><td>950</td><td>18</td><td>17100</td></tr>
<tr><td>Z</td><td>100</td><td>25</td><td>2500</td><td>870</td><td>27.5</td><td>23925</td></tr>
</table></td><td>5</td></tr>
<tr><td>CO- 2 ; SO- ; BL- Apply</td><td>d</td><td>The annual demand for a product is 6,400 units. The unit cost is ₹ 6 and inventory-carrying cost per unit per annum is 25% of the average inventory cost. If the cost of procurement is ₹ 75. Calculate:
(a) Economic order quantity,
(b) Number of orders per annum, and
(c) Time between two consecutive orders.</td><td>5</td></tr>
</table>

Q.2 CO-2 ; SO-; BL- Apply	a	From the following details furnished by Mittal Ltd. for the year ended 31st March 2022, prepare a Cost Sheet showing the elements of cost also find out the profit for the year. **Particulars** / **Rs.** Salesman Salary — 45000 Opening stock of Finished goods (2000 units) — 560000 Directors fees — 97800 Indirect wages — 345000 Repairs to office furniture — 40500 Works Managers salary — 109700 Showroom expenses — 11300 Depreciation on Computer — 1200 Indirect Materials — 630000 Depreciation on Plant and Machinery — 27500 Advertisements — 153000 Office Salary — 790000 Direct wages — 20200 Direct materials — 97500 Direct expenses — 32000 Closing stock of Finished goods (2000 units) — ---------- Other information : 1. Closing stock of Finished goods is to be valued at cost of production 2. Profit desired on sales is 20% 3. Number of units sold during the year was 10000	15
Q.2 CO-2; SO-; BL- Understand	b	Explain the benefits of good inventory control system for an organization.	5

<table>
<tr>
<td>Q.3
CO-2; SO-;
BL-Apply</td>
<td>a</td>
<td>For the manufacture of a product X, two components A and B are used.
The following particulars about the components are:
Normal usage (per week) 300 nos.
Maximum usage (per week) 450 nos.
Minimum usage (per week) 150 nos.
Reorder quantity A-2400 nos. , B-3600 nos.
Reorder period A-4 to 6 weeks, B- 2 to 4 weeks
You are required to calculate the following for both the components.
a) Reorder quantity
b) Maximum level and
c) Minimum level.</td>
<td>8</td>
</tr>
<tr>
<td>Q.3
CO-3 SO-;
BL-
Understand</td>
<td>b</td>
<td>Given are the information of XYZ Ltd.
<table>
<tr><td>Particulars</td><td>X</td><td>Y</td></tr>
<tr><td>Raw material per unit</td><td>160</td><td>120</td></tr>
<tr><td>Wages</td><td>80</td><td>60</td></tr>
<tr><td>Selling price per unit</td><td>500</td><td>400</td></tr>
</table>
Fixed cost is Rs.5,00,000 and variable overheads are 150% of wages.
Alternative sales mix:
a. 3500 unit of X and 2500 units of Y.
b. 6000 units of X and Nil units of Y.
c. 4000 unit of X and 2000 units of Y.
From the above information, you are required to find out the most profitable sales mix.</td>
<td>12</td>
</tr>
<tr>
<td>Q.4
CO- 3; SO-;
BL-Apply</td>
<td>a</td>
<td>A factory is currently running at 60% capacity and produces 6000 units The cost per unit details is given below:
Material Rs.15
Labour Rs. 4
Factory overheads Rs. 15 (Rs. 6 fixed)
Administrative overheads Rs.5 (Rs. 3 fixed)
The current selling price is Rs. 50 per unit. At 70% working, material cost per unit increases by 5% and selling price per unit falls by 2%. At 90% working, material cost per unit increases by 8% and selling price per unit falls by 5%. Prepare a flexible budget showing profits of the factory at 70% and 90% working and offer your comments.</td>
<td>15</td>
</tr>
</table>

Q.4 CO- 4; SO-; BL- Evaluate	b	Explain FIVE applications of Marginal costing for taking various types of managerial decisions.	05
Q.5 CO-5; SO-; BL-Apply	a	Ambuja Garments produces two products "SUCHI" and "RUCHI" which are popular in market. The management has four different options to produce both the products, given as under. **Particulars** / **Suchi (Units)** / **Ruchi (Units)** Option I: 800 / 600 Option II: 1,600 / -- Option III: -- / 1,300 Option IV: 1,100 / 500 **The per unit production cost/sale data are :** **Particulars** / **Suchi** / **Ruchi** Direct Material (₹): 25 / 30 Direct Labour (Hours): 10 / 12 Variable Factory Overheads is 100% of direct Labour cost for both products Selling Price (₹): 75 / 90 Labour Rate is ₹ 2 per hour Total Fixed Overheads for both products ₹ 10,000 You are required to evaluate the options and identify the most profitable sales-mix.	15
Q.5 CO-2; SO-; BL- Evaluate	b	The standard material cost for 100 kg of a product Coke is made up of : Material A- 30 kg @ Rs 4 per kg Material B- 40 kg @ Rs.5 per kg Material C- 80 kg @ Rs. 6 per kg A batch of 500 kg of product Coke is made up of : Material A – 140 kg at a total cost of Rs. 588 Material B- 220 kg at a total cost of Rs. 1056 Material C- 440 kg at a total cost of Rs. 2860 Calculate all Material Variances.	5

Q.6 CO- 4, SO-; BL- Evaluate	a	Following information is available in respect of G Ltd. And D Ltd: **Particulars** / **G Ltd. (Rs)** / **D Ltd. (Rs)** Sales 11,00,000 14,00,000 Variable cost 8,80,000 10,50,000 Profit 1,20,000 2,00,000 **Calculate:** i P/V Ratio of both companies ii. Fixed cost of both companies iii. Break-even point of both companies iv. Sales to earn profit of Rs 2,10,000 by each company v. Margin of safety of 'D' Ltd.	15
Q.6 CO- 1; SO-; BL- Remember	b	Explain any FIVE limitations of Financial Accounting.	5
Q.7 CO- 1; SO-; BL- Remember	a	Write Short Notes on: 1. Management Accounting 2. Activity Based Costing (ABC) 3. Importance of Budgeting	15
Q.7 CO- 4; SO-; BL- Evaluate	b	Using the following information, calculate labour cost, rate , efficiency variances : The budgeted labour force for producing product X is : 20 Semi-skilled workers @ 75 paise per hour for 50 hours 10 Skilled workers @ Rs 1.25 per hour for 50 hours The actual labour force employed for producing X is : 22 Semi-skilled workers @ 80 paisa per hour for 50 hours 8 Skilled workers @ Rs 1.20 per hour for 50 hours	5

THE END